



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6122**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	I	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600010
Course Section:	1				

L61-468_986_1_1505_48258



Name of the candidate. Chahat Dholakia

Candidate's UID number:

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Examination class room number:

2	0	2
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Invigilator's signature with date:

	08/10/2025
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(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

PROGRAM: CSE - AIML

DAY AND DATE: Wednesday 8th Oct 2025

COURSE NAME: Deep Learning

SEMESTER: VII

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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Q1 a)

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

Gradient update rule states that

$$w_{new} \leftarrow w_{old} - \eta \frac{\partial f(w)}{\partial w}$$

$$\frac{\partial f(w)}{\partial w} \Rightarrow \frac{\partial}{\partial w} w \|Xw - y\|$$

$$= w \|Xw - y\|$$

$$w_{new} \leftarrow w - \eta (w \|Xw - y\|)$$

for convergence

$$\eta < \frac{2}{\lambda_{max}}$$

on calc

we calculate eigen values using

$$\frac{\partial^2 f(w)}{\partial w^2}$$

$$\left[\right]$$

→ matrix.

$$\eta < \frac{2}{\lambda_{max}} \text{ to get convergence}$$

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1 b) Square error = $\frac{1}{2} | \hat{y} - y |^2$

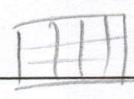
 $\hat{y} \rightarrow$ predicted output $y \rightarrow$ expected output

Overfitting occurs when model is trained too much on training data however, when tested on test data it performs poorly. This happens because model is trained more than necessary and thus when given some new data begins to hallucinate, and hence it affects generalization. Since the model is not able to generalize.



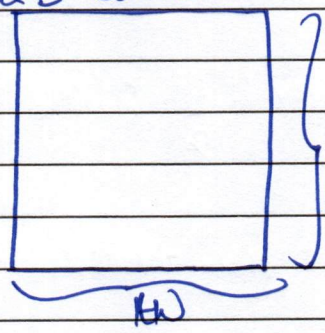
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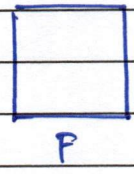


2 a)

2D convolution.



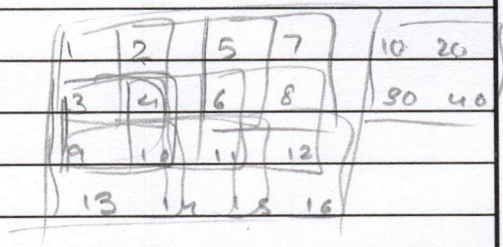
H



F

F

F=2



$((1 \times 10) + (20 \times 2) + (30 \times 3) + (40 \times 4)) \div 3 (4 \times 3)$

Assuming $\rightarrow$ No padding } standard
stride = 1

Suppose $H = W = N$

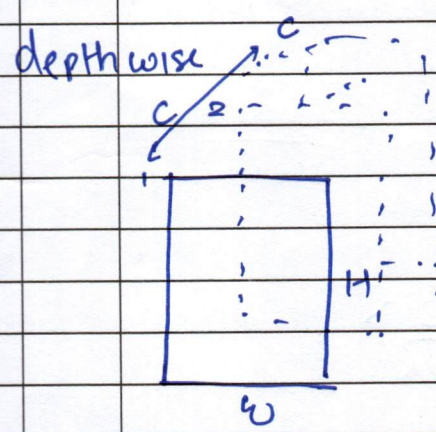
Assume

At 1st it will perform $F \times F \rightarrow$ computation.
Now depending on N it will perform $N-1$ times
 $(N-1) \times (F \times F)$

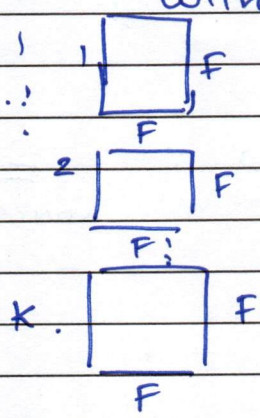
this it performs again $(N-1)$ times for entire to complete

Hence $(N-1)((F \times F) \times N-1)$

Hence the new matrix is of size
 $(N-F+1) \times (N-F+1)$



With K filters of $F \times F$



This

It will give K output images. with

$(H-1 \times F) \times (W-1 \times F) \times C \times K$

Sr. No.

6122



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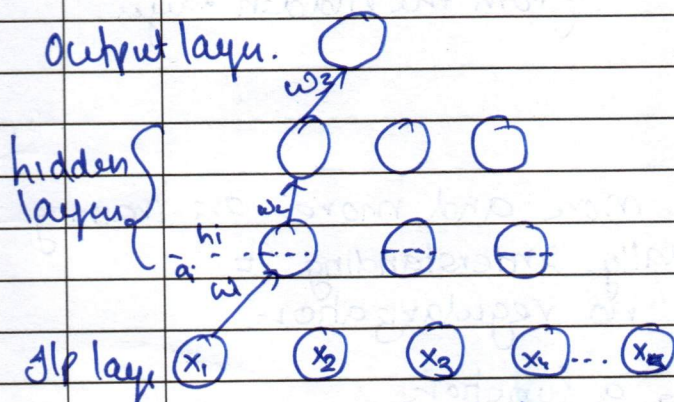
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2 6

$$MSE = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

$\rightarrow$ No. of observations.

Undercomplete autoencoder $\rightarrow$ when dimension of hidden size $<$ dim of i/p layer.



Now as the i/p goes into the hidden layer and hence its size is reduced. Thus in Undercomplete auto encoder it is able to incorporate all the i/p in a reduced form.

$\rightarrow$ Sigmoid function.

$$a_i = (x_i w_i + b)$$

$$h_i = \sigma(a_i)$$

Output function

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - x_i w_i)^2$$

$\downarrow$
Sample space (inputs)

$$\frac{\partial L}{\partial w} = -\frac{2}{N} \sum_{i=1}^N (y_i - x_i w_i) x_i$$

$$= -\frac{2}{N} \sum_{i=1}^N (x_i) (y_i - x_i w)$$

Since the i/p size is reduced it is similar to principal component analysis. Since the size even in PCA it only takes the imp stuff from the input reducing its size.

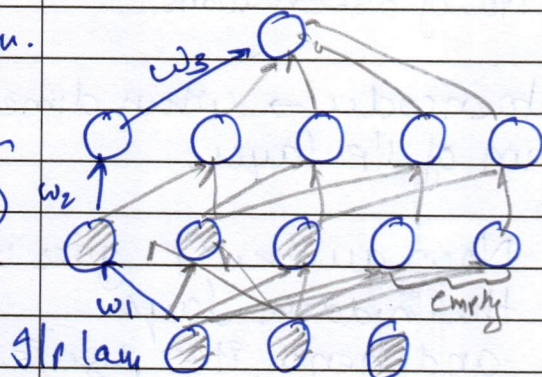
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An overcomplete autoencoder.
the dim of hidden layer is more than
dim of input layer.

o/p layer.

Hidden
Layer

In this as we see
that since hidden layer
has more dimension
it retains everything
from the hidden layer.

hence it only learns more and more on training
data without actually understanding it
and hence there is no regularization.

$$Y_i = \sum w_i x_i + b. \rightarrow \text{a function.}$$

$$\text{final output} = f(Y_i)$$

now since it's only retaining the i/p value
without understanding the ^{out} function for final
output it tends to learn identity function.

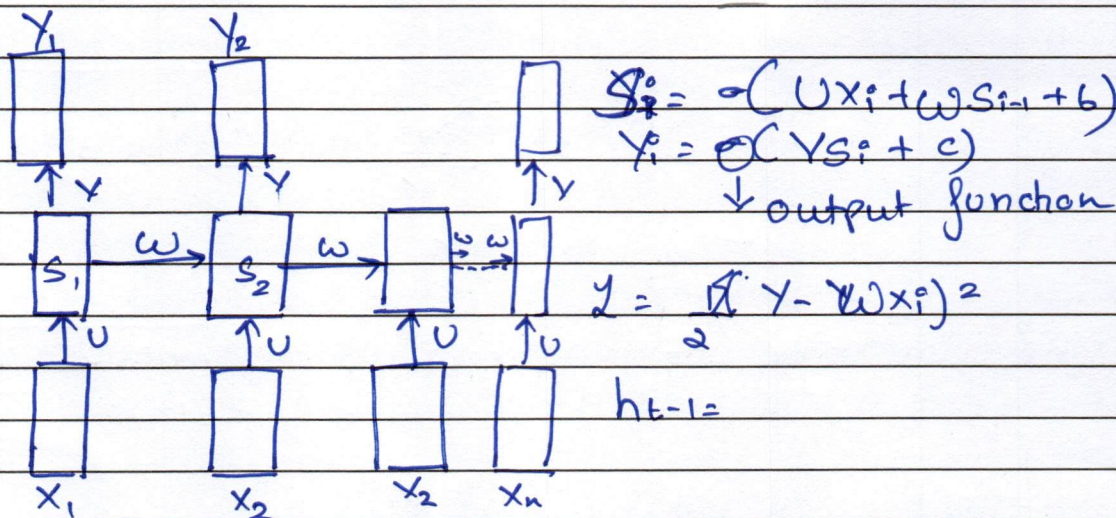
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2L

3 a.



$$\frac{\partial L}{\partial w}$$

$L \rightarrow Y \rightarrow s_i \rightarrow w$
dependency.

 $h_{t-k} \Rightarrow s_{t-k}$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial Y} \times \frac{\partial Y}{\partial s_i} \times \frac{\partial s_i}{\partial w}$$

$$\frac{\partial L}{\partial Y} \text{ or } \frac{\partial L}{\partial h_{t-k}} \Rightarrow$$

forget gate in LSTM helps modifying recursion since it discards values not required

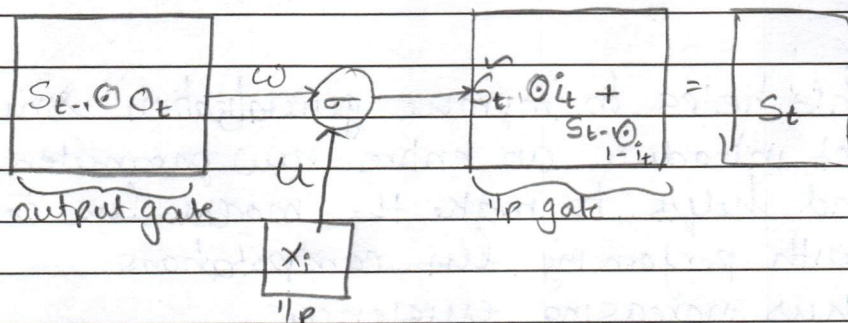
$$s_t = \sigma(w_s h_{t-1} \cdot U_s x_t + b)$$

With this mathematical equation

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3 b



Unlike in LSTM GRU only have 2 gates
 Output / update gate $\rightarrow$ to ~~keep~~ what information to retain
 Input / reset gate $\rightarrow$ what not to retain.

gates equation

$$O_t = \sigma(W_o S_{t-1} + U_o x_t + b) \quad [\text{Output gate}]$$

$$I_t = \sigma(W_i S_{t-1} + U_i x_t + b) \quad [\text{Input gate}]$$

S_{t-1} is the temporary output of previous state.

States equation

$$\tilde{S}_t = \sigma(W(S_{t-1} \odot O_t) + U x_t + b)$$

$$S_t = \tilde{S}_t \odot I_t + S_{t-1} \odot (1 - I_t)$$

$\hat{I}_t \rightarrow$ input at time t
 $\tilde{S}_t \rightarrow$ temp output at time t

here as we can see there is no explicit forget gate hence there is one less parameter.

where there will be one more gate eg in LSTM
 forget gate $S_t = \sigma(W_f h_{t-1} + U_f x_t + b)$

$$h_t = S_t \odot O_t$$

and also

$$S_t = \tilde{S}_t \odot I_t + S_{t-1} \odot (1 - I_t)$$

$\rightarrow$



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MarksQuestion
No.

This helps to improve generalization since we need not introduce an entire new parameter and helps to make the model less complex with performing less computations. Thus increasing efficiency

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Sr. No. 6122



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Question
No.

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[illegible]

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